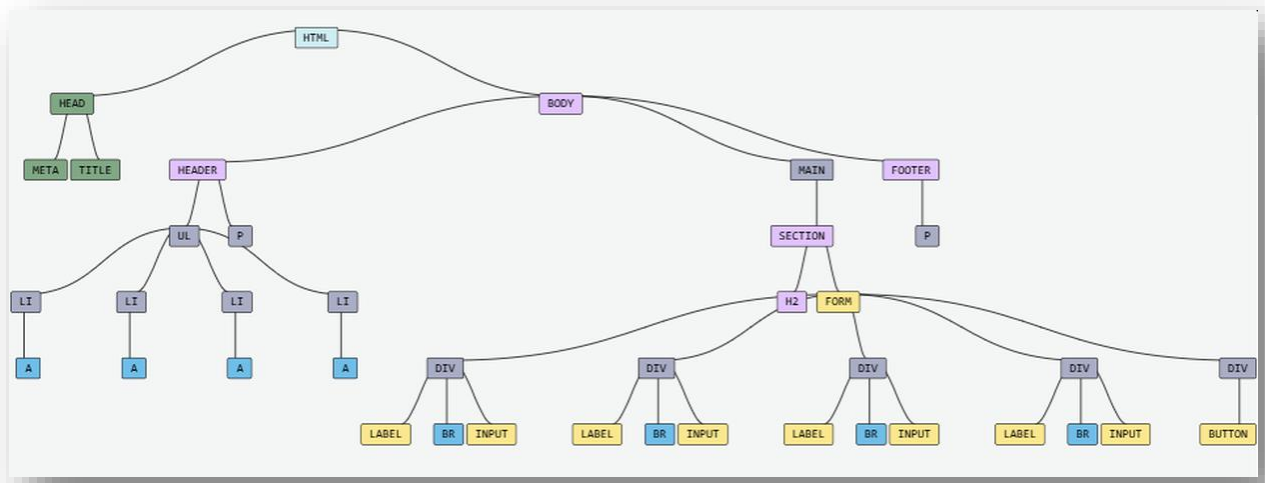
Simple Sign-Up Web Page

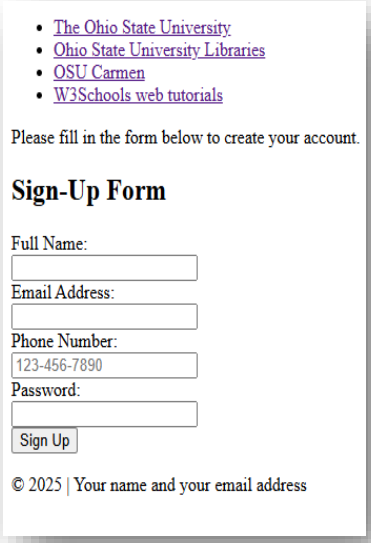


HTML Document Object Model (DOM) tree

Part I

Design an HTML page that allows a user to create an account. Your page should include the following:

1. **Page Structure**
   - ❖ Use <!DOCTYPE html> and include <html>, <head>, and <body> tags.
   - ❖ Set the page language to English and provide a title.
2. **Header**
   - ❖ Include a list of at least four useful links, such as:
     - A university homepage
     - A university library
     - A learning platform (e.g., Carmen)
     - A tutorial or learning website (e.g., W3Schools)
   - ❖ Add a short paragraph inviting users to fill in the form.
3. **Main Section**
   - ❖ Include a heading for the sign-up form.
   - ❖ Add a <form> that uses the POST method.
   - ❖ Include the following input fields (all required):
     - Full Name (text)
     - Email Address (email)
     - Phone Number (tel) with a placeholder format
     - Password (password)
   - ❖ Add a submit button labeled “Sign Up”.
4. **Footer**
   - ❖ Include a footer with your name, email, and the current year. Use semantic HTML tags such as <header>, <main>, <section>, and <footer>.



Part II

- Create a folder called css and place the attached responsive.css file inside it for this activity.
- Add the following tags to your HTML file:
  1. <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  2. <link rel="stylesheet" href="css/responsive.css">

- Move the * Please fill in the form below to create your account. * from the <header> tag to the <main> tag.

```
<main>
    <p>Please fill in the form below to create your account.</p>
    <h2>Sign-Up Form</h2>
    <form action="" method="post">
        <div>
```

- In your HTML file:
  1. Add this paragraph element:
  2. Place your cursor between the opening and closing tags.
  3. Type lorem120 and press Enter (or Tab, depending on your VS Code setup).
  4. VS Code will automatically generate 120 words of Lorem Ipsum text inside that paragraph, something like this:

```
</main>

<p class="sidebar">
    Lorem ipsum dolor sit amet consectetur, adipisicing elit. Ullam magnam, reiciendis e
    doloribus animi autem iste dolore esse, earum recusandae culpa magni maiores aperiam
    corporis culpa dolores provident at cumque ut porro minus consequatur ducimus quae n
    laudantium suscipit, doloribus quos quisquam id neque consectetur voluptate vel numq
    facilis, quaerat, iusto iste, sed facere est aliquam eaque libero. Omnis ad officia
    voluptatem! Molestiae, laboriosam at. Autem enim consequuntur quidem illum veritatis
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    error corrupti mollitia beatae veritatis. Deserunt non nostrum doloremque voluptates
    exercitationem cupiditate, nihil, error blanditiis quis, quidem dignissimos?
</p>
```
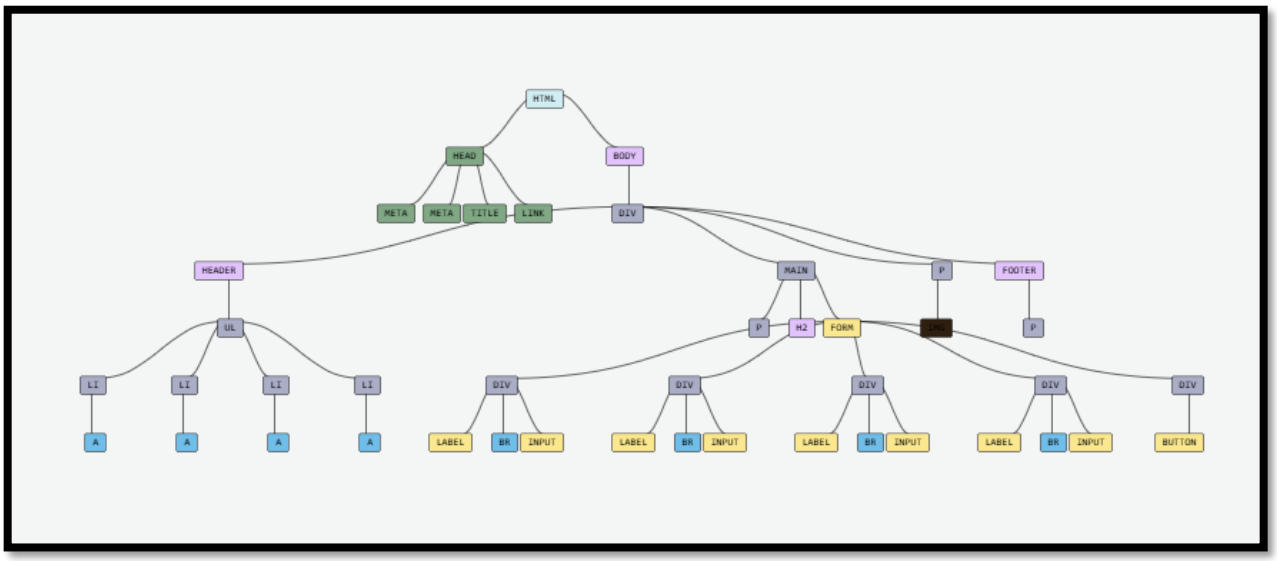
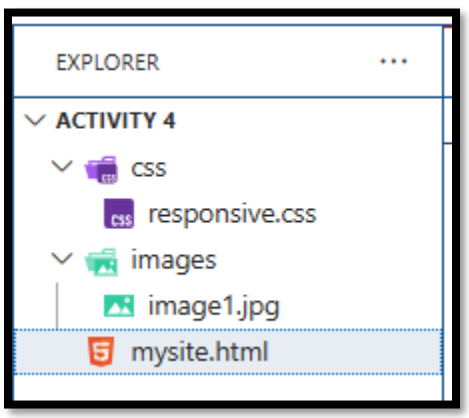
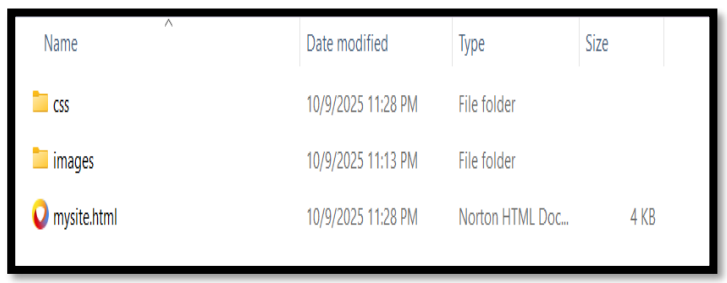

- Visit the following site https://unsplash.com/t/nature , download any image, then create a folder called images and save the image in it. Add the following image tag inside the above element: <img src="images/image1.jpg" class="my-image" alt="Sample image">

- Your DOM structure should look like the diagram below.



- Your working tree should look like the images below.





Part III

- Study the responsive.css file and complete the missing code. For this activity, there is a video you can watch to understand the final design. You are required to create layouts for mobile, tablet, and desktop screens based on the screen resolution.
- In general, you must create a tag and place your <body> content inside it. Then, apply the wrapper class.

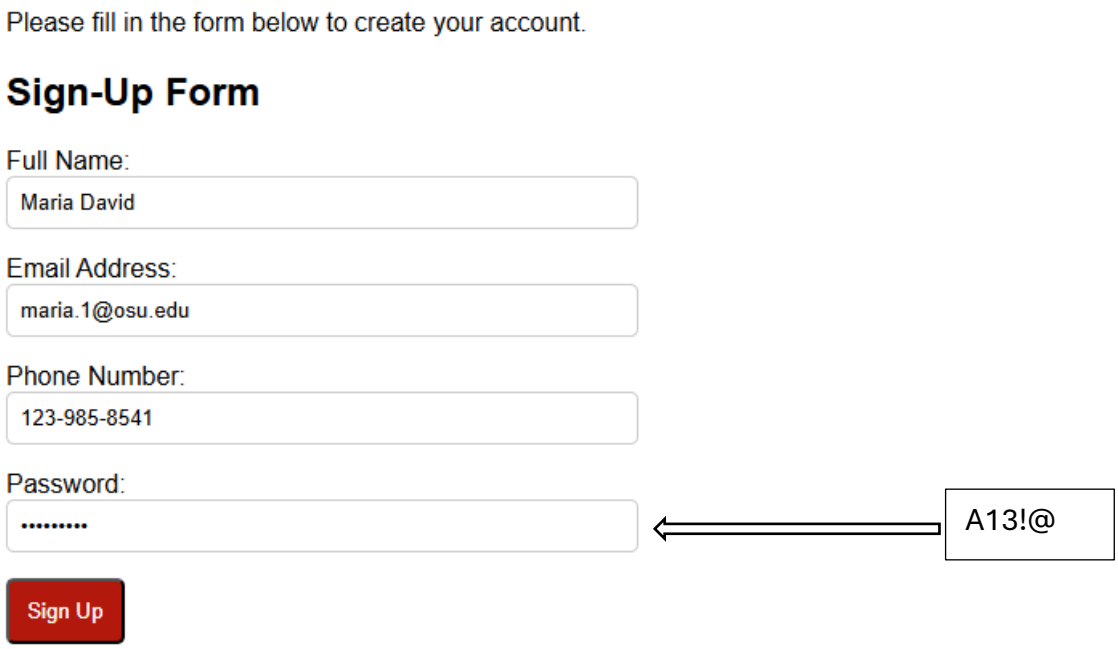
```
<body>
    <div class="wrapper">
        <header class="top">
            <ul class="menu">
                <li><a href="https://www.osu.edu">The Ohio State University
```

- Apply the footer class to the <footer> tag.

```html
            exercitationem cupiditate, nihil, error blanditiis quis, quidem dignissimos?
            <img src="images/image1.jpg" class="my-image" alt="Sample image">
        </p>

        <footer id="footer">
            <p>&copy; 2025 | Your name and your email address</p>
        </footer>

    </div>
</body>
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
   ..............................................
</head>
<body>
  <div class="wrapper">
    <header class="top">
      <ul class="menu">
        ...................
      </ul>
    </header>

    <main>................
      <form .....>
          ......
        <div>
          <button type="submit">Sign Up</button>
        </div>
      </form>
    </main>
    <p class="sidebar">
      Lorem ipsum dolor sit amet consectissimos? .....
      <img src=............................>
    </p>
    <footer id="footer">
      .........................
    </footer>
  </div>
</body>
</html>
```

Part IV



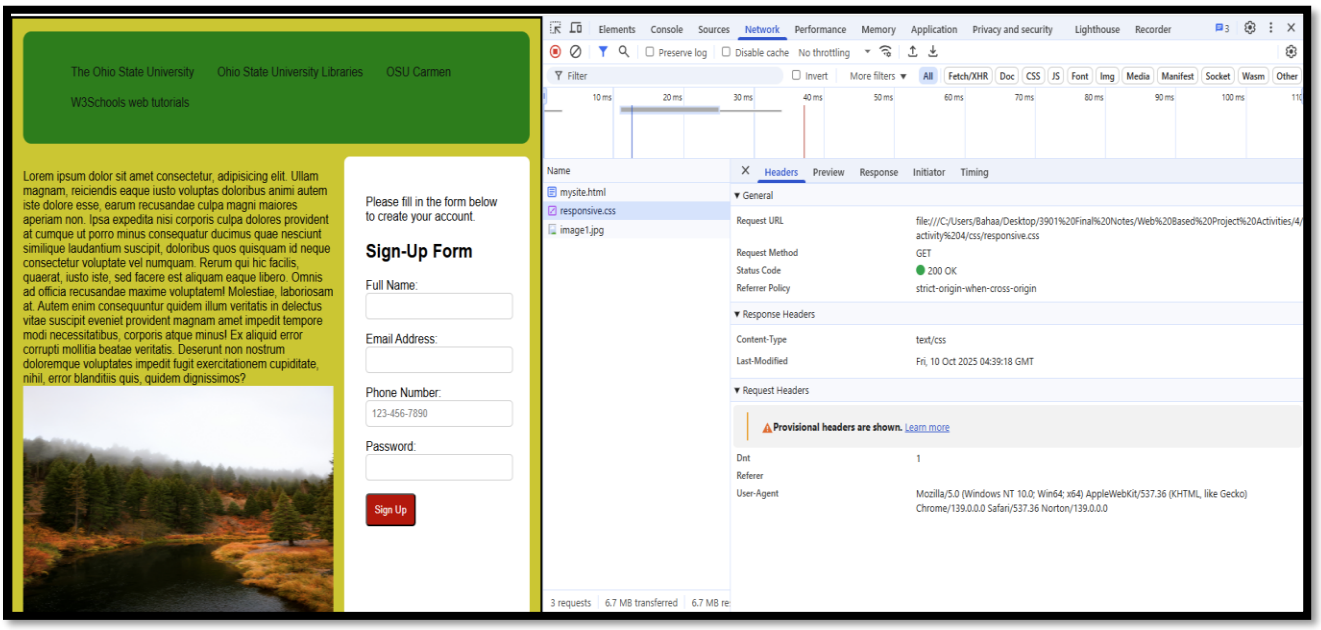
In the context of the HTTP protocol, write two HTTP requests related to the image above. Assume the requests use HTTP/1.1. One request should use the GET method, and the other should use the POST method. assume the password length is 5 characters(a13!@).

- **GET Request:**
  - Include the request line with the host www.osu.edu and the method GET.
  - Include the required HTTP headers, including any meta headers.

- **POST Request:**
  - Include the request line with the host www.osu.edu and the method POST.
  - Include the required HTTP headers, including any meta headers.
  - Include the body with the appropriate Content-Length.

For your answer, create a web page named HTTP and include a link to it in the header section. Submissions in the form of notes, Word documents, PDFs, or any other file types will not be accepted.

Part V



- When you open the webpage mysite.html in your browser (as shown in the image), the browser fetches several resources to render the page.
- Explain the sequence in which the browser retrieves the files: HTML, CSS, and image based on the Network tab shown in the Developer Tools.
- Why does the browser need to fetch the HTML file first before the CSS and image files?
- From the Network tab, what does the Status Code [200 OK] mean for the responsive.css file request?

For your answer, create a web page named Browser and include a link to it in the header section. Submissions in the form of notes, Word documents, PDFs, or any other file types will not be accepted.

Instructions for Activity 4:

1. Submit a **.zip file** containing all your work for Activity 4.